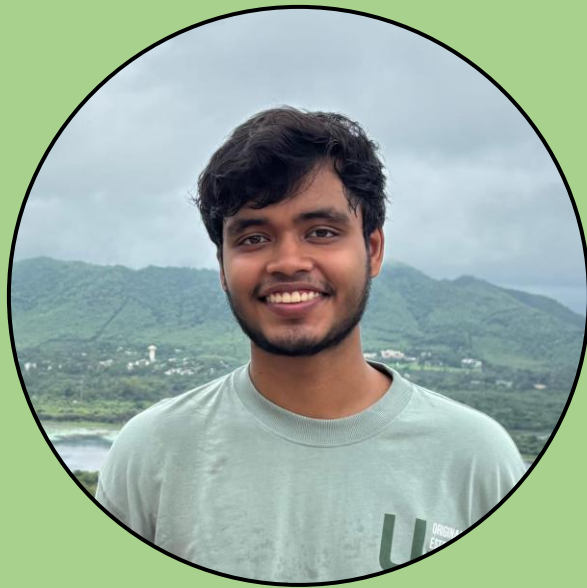


SUZUKI'S IGNITERS INNOVATION



HYDROGEN
VEHICLE
CONCEPT

MEET THE TEAM



ABHIK PAL



ARHAAN KOCCHAR



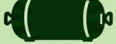
LAKSHMANAN GANAPATH

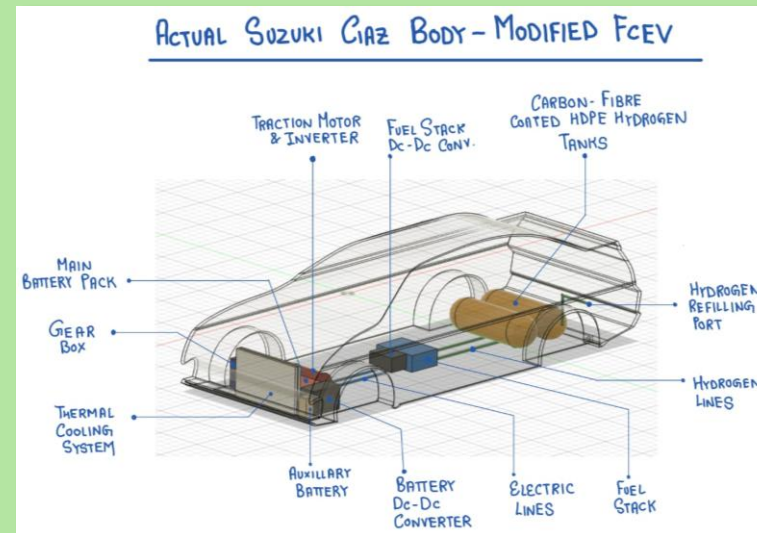
FCEV vs BEV

PARAMETERS	FCEV	BEV
 RANGE	~500 Km (Roughly Same)	~500 Km (Mid Range Sedan BEVs)
 REFUELING / CHARGING TIME	~4.1 min (H ₂ refuel at 1.2 kg/min) Comparable to conventional refueling	~45 min (DC fast charge: 10–80%) ~8–12 hr (Home AC charge: 7–11 kW)
 TOTAL OWNERSHIP COST (8 YEARS / 120000 KM)	Rs. 21.79 / Km (Current estimated, pilot scale) *Reduces with scale, localisation & component maturity	Rs. 13.47 / Km (Mid-size sedan segment) Lower due to simpler powertrain & mature supply chain
 ENERGY EFFICIENCY (Tank/Well to Wheel or Wall to Wheel)	~40% (Well-to-wheel) Losses in H ₂ production, compression, transport & fuel cell conversion	~70% (Wall-to-wheel) Higher overall efficiency from grid to wheels
 INFRASTRUCTURE REQUIREMENTS	Hydrogen production, storage, transport & dispensing stations High initial capex (₹6–15 Cr per station) Few stations currently; corridor-based rollout required	Charging stations (AC & DC) Lower capex per station Widespread grid availability; easier scalability
 SUSTAINABILITY	Zero tailpipe emissions (H ₂ → H ₂ O) Sustainability depends on green hydrogen production (electrolysis using renewable energy)	Zero tailpipe emissions Sustainability depends on clean electricity mix & battery recycling

SAFETY

Each risk maps to a specific, already-proven engineering response used in Type-IV tank and PEMFC practice worldwide:

HAZARD	DESIGN RESPONSE	WHY IT MATTERS
 STORAGE	Type-IV CF tanks, PRDs, controlled venting	Survives fatigue, impact & fire exposure without rupture
 CRASH	Tanks between crumple zones, reinforced frame	Impact energy absorbed before reaching H ₂ storage
 LEAK	H ₂ sensors + automatic crash-triggered shut-off	Light, odourless gas — detection replaces smell
 FIRE	Upward-venting PRDs, heat-resistant fuel lines	Controlled release prevents tank over-pressure rupture
 STACK	Humidity/temp/pressure control, filtration	Prevents membrane drying, flooding & catalyst damage
 HV	Insulated cabling, auto-disconnect relays	Stack runs at hundreds of volts — isolates on fault



Self Made CAD Model of FCEV Concept

NOTE

Kindly refer to annexures for detailed analysis of the listed parameters and calculations

INFRASTRUCTURE

1 CHG INTEGRATION IN CNG PUMPS
Equipment: On ground Hydrogen Storage tanks
Costs about ₹8–15 Cr after Government subsidies
Subsidies vary from state to state and are roughly around ₹4Cr

2 COMPRESSORS
2 Stage compression (300bar → 550bar → 700bar).
Storage in Cascade System: Tanks arranged in a line
CNG also have compressors → Reduces maintenance costs

3 IN HOUSE ELECTROLYSER
Water Electrolyser requires 48–55 kWh of electricity per kg of H₂
For Green Fuel, Solar + Hybrid Grid is required (0.42MW for 2 hours) and costs ₹1.8–4.4 Cr

4 SAFETY & MAINTENANCE
H₂ fuels are highly combustible and pose a heavy threat to surroundings. Quality checks are done by PESO and HAZOP in surrounding 500m.
Maintenance training and equipment cost upto ₹ 13 lakh/year

BUSINESS MODEL

1 TARGET HIGH-UTILIZATION CUSTOMERS
Begin with airport/commercial taxi fleets, government & institutional fleets, and long-distance intercity vehicles that travel high daily distances, where rapid hydrogen refueling resulting in long range travel provide the greatest operational advantage.

2 BUILD H₂ CORRIDORS – NOT NATIONWIDE NETWORK
Use toll-road traffic & GPS mobility data to identify India's highest demand long-distance corridors, major transport/trade routes, and airport–city/intercity routes. Deploy first H₂ stations strategically along these corridors to maximize station utilization & fleet access.

3 SCALE INFRASTRUCTURE CORRIDOR-BY-CORRIDOR
Upon satisfactory vehicle demand, station utilization and economic viability in one corridor, then expand to the next corridor instead of large upfront nationwide infrastructure investment. This creates a progressively expanding H₂ network while improving ROI.

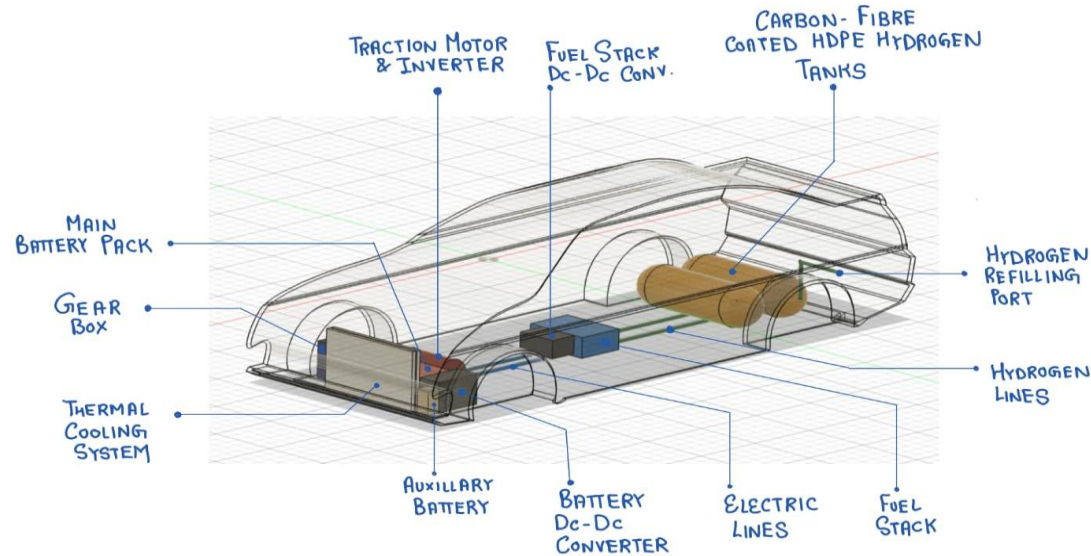
4 DIVERSIFY THE H₂ ECOSYSTEM
Use the established corridors as common CHG pumps for introducing FCEV buses, cargo carriers and other HMs. This increases hydrogen station utilization and spreads infrastructure costs across multiple vehicle segments creating a stronghold in the market in the long run

ANNEXURES

1.  Vehicle Concept and CAD
2.  MATLAB Vehicle Sizing
3. H_2 Hydrogen Storage
4.  Fuel Stack
5.  Energy Efficiency
6.  Performance & Power
7.  Total Cost of Ownership
8.  Safety Architecture
9.  Existing Hydrogen Infra.
10.  Manufacturing
11.  Hydrogen Infra. Costs
12.  Sourcing & Assumptions

ANNEXURE 1 – VEHICLE CONCEPT & CAD

ACTUAL SUZUKI CIAZ BODY – MODIFIED FCEV



[CLICK HERE FOR THE CODE](#)

COMPONENT PLACEMENT

- 2 × Type-IV H₂ tanks — rear underfloor, between crumple zones
- 55 kW fuel-cell stack — vehicle center, rigid frame + isolators
- Buffer battery + DC-DC converter — front compartment
- Traction motor + inverter, gearbox — front, existing EV mounting
- H₂ lines routed low, away from cabin and HV runs
- Refilling port — rear, external access
- Thermal cooling system — shared loop, stack + battery

This is a self-modelled concept-study CAD layout used to validate component packaging against a real vehicle body — not a manufacturing drawing. Dimensions are taken from the physics-based sizing tool (Annexure 2), not assumed.

ANNEXURE 2 – MATLAB VEHICLE SIZING: SOURCE OF TRUTH

CONVERGED OUTPUT OF THE PHYSICS-BASED SIZING TOOL (ROAD-LOAD, TANK GEOMETRY, STACK SIZING, DRIVE-CYCLE SPLIT)

Metric	Value	Confidence / basis
Range	500 km (design target)	Sourced — sizing-tool input
Total vehicle mass	1,603 kg (1,620 kg w/ correction)	Concept estimate
Usable H ₂ stored	4.95 kg total (2.48 kg × 2 tanks)	Sourced — MATLAB output
H ₂ consumption	0.99 kg / 100 km	Sourced — MATLAB output
Tank OD / overall length	341.5 mm / 1,048.6 mm each	Sourced — MATLAB output
Tank internal volume (total)	128.2 L	Sourced — MATLAB output
Tank shell mass (total)	62.06 kg (52.2 CF + 9.8 liner)	Sourced — MATLAB output
Liner / CF overwrap thickness	5.0 mm / 15.75 mm	Sourced — MATLAB output
FC stack rated power	55 kW net	Sourced — MATLAB output
FC stack cells / voltage / area	462 / 300.3 V / 152.6 cm ² per cell	Sourced, cross-checked
FC stack length / mass / Pt	964 mm / 27.5 kg / 24.7 g	Sourced — MATLAB output
Buffer battery	3.36 kWh, 37.3 kg (NMC811)	Corrected value
Traction motor	38.5 kg, sized to 126.9 kW peak	Sourced — MATLAB output
Governing power case	Accel (126.9kW) > climb (44.9kW) > cruise (21.4kW)	Sourced — MATLAB output
Tank-to-wheel efficiency (cruise)	40.6%	Sourced — MATLAB output

Modelling limitation: corrected buffer battery (37.3 kg) not fed back into mass-convergence loop – used consistently here; +~1% mass, no conclusion changes.

TANK GEOMETRY (PER TANK, 2× FITTED)

Parameter	Value
Outer diameter	341.5 mm
Overall length (incl. domes)	1,048.6 mm
Internal volume	64.1 L (128.2 L total)
Liner (HDPE) thickness	5.0 mm
CF overwrap thickness	15.75 mm (75.9% of shell)
Shell mass	31.0 kg / tank (62.06 kg total)
Usable H ₂ per tank	2.48 kg

STORAGE SATEFY DESIGN

- CF-reinforced composite overwrap over HDPE liner — withstands high internal pressure, impact and fatigue
- Positioned behind passenger compartment, within reinforced frame; spread front-to-back between crumple zones

MONITORING & RELIEF

- Continuous pressure and flow monitoring — prevents cell voltage reversal from H₂ starvation
- Pressure-relief devices vent upward and away from occupants in a fire event, preventing tank rupture

STACK SIZING (CONVERGED, CROSS-CHECKED)

Parameter	Value
Rated power	55.0 kW net
Cell count / stack voltage	462 cells / 300.3 V (183.2 A)
Active area per cell	152.6 cm ² (~143×143 mm plate)
Stack length	964 mm (462 cells + 2 end plates)
Total platinum loading	24.7 g
Stack mass	27.5 kg
Cell-count sizing cross-check	30.7 kg (independent method)
Specific-power sizing cross-check	27.5 kg (independent method)

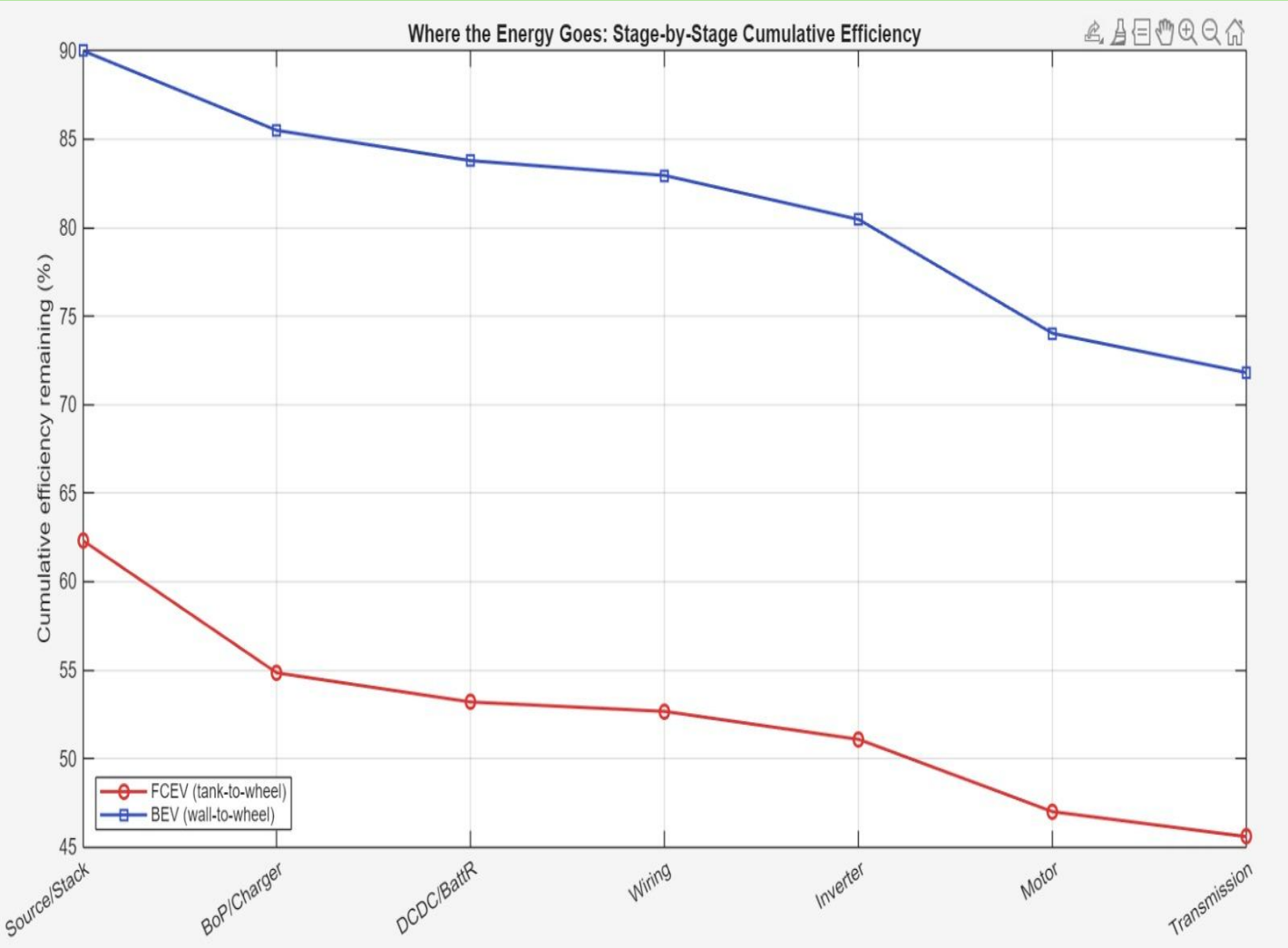
MEMBRANE & WATER MANAGEMENT

- Humidity, temperature & pressure control prevent drying, flooding, fatigue, contamination
- Reinforced-polymer membrane resists cracking; flow channels + air compressor remove excess product water

ELECTRICAL & CONTAMINANT PROTECTION

- Insulated HV cabling, insulation monitoring, auto-disconnect relays, waterproof connectors
- Air filtration excludes CO, sulfur, ammonia — protects catalyst activity

FCEV'S EFFICIENCY PENALTY IS THE PRICE OF RAPID REFUELLING — NOT A CASE AGAINST HYDROGEN



WHERE FCEV LOSES ENERGY

- Stack (chemical→electrical) is the largest loss stage — a reforming/conversion penalty BEV never incurs
- BoP, DC-DC, wiring, inverter, motor, transmission losses are broadly comparable to BEV

HEADLINE NUMBERS

- FCEV tank-to-wheel: ~40% (40.6% at cruise)
- BEV wall-to-wheel: ~70% (71.8% incl. AC charging)
- BEV ~1.7–1.8× more efficient at same load point

HONEST FINDING — EFFICIENCY GAP

FCEV is not more efficient than BEV — it is less efficient by a wide, structural margin. The strategic argument is not that this gap doesn't matter; it's that a segment paying for uptime and refuel speed can absorb it, while a mass-market buyer generally cannot.

DRIVE-CYCLE POWER REQUIREMENT

Case	Power required
120 km/h cruise	21.44 kW
0–100 km/h in 11 s (governing)	126.86 kW
30° grade climb @ 20 km/h	44.88 kW
Motor peak-power requirement	126.86 kW

SOURCE SPLIT

Component	Rating
Fuel-cell stack (continuous)	55.0 kW
Battery buffer (transient)	~75 kW / 3.36 kWh @ 30C
Traction motor + inverter (peak)	126.9 kW

WHY TWO POWER SOURCES

- Acceleration, not cruise or climb, sets the peak-power requirement — a short, high-power transient
- Sizing the fuel cell alone to 127 kW would make the stack far larger, heavier and costlier than necessary for a load it only sees briefly
- A small power-oriented buffer battery covers the transient; the stack is sized to continuous/cruise-class load instead

HONEST FINDING – DESIGN IMPLICATION

Motor and power electronics must be priced and packaged as ~127 kW peak-rated components, not 55 kW ones — a direct cost driver flagged in the manufacturing annexure.

TCO COMPARISON (8 YEARS / 120000 km)

COST COMPONENT	FCEV (Rs)	BEV (Rs)
 PURCHASE COST	2,800,000	1,800,000
 FUEL / ENERGY COST	475,356	170,494
 MAINTENANCE COST	180,000	96,000
 RESALE VALUE	-840,000	-450,000
 TOTAL TCO (8 YEARS / 120000 km)	2,615,356	1,616,494
 TCO PER KM	21.79 Rs/km	13.47 Rs/km

IMPORTANT NOTE

As Suzuki moves to a larger scale production, the manufacturing cost per unit reduces and thus total cost of ownership also reduces with time

- Ownership period: 8 years | Annual usage: 15,000 km/year | Total distance: 120,000 km
 - FCEV purchase: ₹28.0 lakh | BEV purchase: ₹18.0 lakh
 - FCEV hydrogen consumption: ≈0.99 kg/100 km | Total H₂ consumed: ≈1,188 kg
 - FCEV refuelling: ≈240 refills over 8 years | ≈4.95 kg/refill | ≈₹1,981/refill
 - Total FCEV fuel cost: ₹4,75,356 at ₹400/kg H₂
- FCEV maintenance: ≈12 scheduled services | ₹15,000/service | ₹1,80,000 total
 - BEV energy cost: ≈21,312 kWh over 120,000 km | ₹8/kWh | ₹1,70,494 total
 - BEV maintenance: ≈8 scheduled services | ₹12,000/service | ₹96,000 total
 - Resale value: FCEV ₹8.40 lakh | BEV ₹4.50 lakh
 - Final TCO: FCEV ₹26.15 lakh (₹21.79/km) vs BEV ₹16.16 lakh (₹13.47/km)

HAZARD	DESIGN RESPONSE	WHY IT MATTERS
High-pressure storage	CF-reinforced Type-IV tanks; continuous pressure & flow monitoring	<i>Withstands pressure/impact/fatigue; starvation-driven voltage reversal prevented, not just detected</i>
Crash	Tanks within reinforced frame between crumple zones; stack on vibration isolators near vehicle centre	<i>Side impacts comparatively safer; crumple zones absorb energy before reaching H₂ or the stack</i>
Leak detection & mitigation	Crash-triggered auto shut-off; sensors monitor tank bay, stack, piping and cabin continuously	<i>H₂ is light, odourless, invisible — detection engineering substitutes for human senses</i>
Fire	PRDs vent upward away from occupants; fire-resistant fuel-line materials delay heat transfer	<i>Controlled venting prevents rupture and buys occupants evacuation time</i>
Fuel-cell stack	Humidity/temp/pressure control; reinforced membrane; water/air management; contaminant filtration	<i>Prevents membrane drying, flooding, fatigue and catalyst contamination — main stack failure modes</i>
High voltage	Insulated HV cabling with monitoring; auto-disconnect relays; waterproof connectors	<i>Stack runs to hundreds of volts — isolation on fault eliminates accessible voltage</i>

Anchor	Implied station cost	Basis / confidence
Bottom-up equipment estimate	₹3.1–6.2 cr	GUESSTIMATE — no independent source
Kerala pilot (per station)	≈₹14.7 cr	SOURCED — ₹29.4cr ÷ 2 stations
Maharashtra subsidy cap (30%)	≈₹15 cr	SOURCED
Gujarat 2025 subsidy cap (30%)	≈₹13.3 cr	SOURCED
GreenH market-reported range	₹10–20 cr	SOURCED — broadest Indian figure
Recommended planning range	₹6–15 cr / 350-bar station	50–260 kg/day, excl. electrolysis/land

IOCL GUJARAT

- 550-bar cascade, on-site H₂ generation; operational into 2026, feeding Ahmedabad–Surat corridor

KERALA ANERT PILOT

- 2×260 kg/day @350 bar, Trivandrum & Kochi; Ashok Leyland trucks, ~90% VGF-funded

NTPC LEH–LADAKH

- India's first operational green-H₂ station: 80 kg/day @350 bar + 1.7 MW solar, 5 buses

PESO CHG DISPENSING LICENCE (FORM H)
[REGULATORY]

- 500 m surrounding-area layout study, HAZOP per IEC 61882
- 2 diagonal staircases, CCTV (15-day retention), 3+ emergency stops

STANDARDS

- IS 16061:2021 fuel quality; IS/ISO 17268:2020 refuelling connections
- BIS aligned to ISO 14687:2025

■ KEY ASSUMPTION

Suzuki already has EV manufacturing infrastructure — BIW/press shop, motor + inverter, battery-pack assembly, electronics, dealer/service network, marketing and core R&D. Only hydrogen-specific infrastructure and components are treated as incremental.

Metric	Stage 1 — Pilot	Stage 2 — Full-scale
Production volume	~50–100 units/year	~100,000+ units/year
Total incremental capex	~₹40–65 L	~₹650–1,050 Cr
Cost per unit	~₹22–35 L	~₹6.1–9.65 L (pre-margin)
Indicative price incl. margin	—	~₹6.5–10.7 L
Main cost drivers	Fuel-cell stack + H ₂ tanks	Fuel-cell stack + H ₂ tanks

Fuel-cell stack and H₂ tanks remain the dominant incremental cost drivers at every scale.

■ STAGE-WISE COST BREAKDOWN

STAGE 1 — PILOT PRODUCTION

- H₂-safe bay retrofit, ventilation & sensors: ₹25–40 L
- Fuel-cell stack integration + H₂ test equipment: ₹15–25 L
 - Total incremental capex: ~₹40–65 L
 - Shared EV components (BIW, motor, inverter, battery): ₹6–9 L
 - Bought-out fuel-cell stack: ₹8–12 L | Bought-out H₂ tanks: ₹2–3 L
 - H₂ lines + integration: ₹0.5–1 L
 - Incremental FCEV R&D: ₹3–6 L | H₂-specific certification: ₹1–2 L
 - Shared SG&A + marketing + dealer allocation: ₹1–2 L
 - Total cost per unit: ~₹22–35 L

STAGE 2 — FULL-SCALE PRODUCTION

- Fuel-cell stack manufacturing plant: ₹500–800 Cr
- H₂ tank winding + autoclave plant: ₹150–250 Cr
- Total incremental capex: ~₹650–1,050 Cr
- Shared BIW, motor, battery & electronics: ₹3–4.5 L
- In-house fuel-cell stack: ₹1.5–2.5 L | In-house H₂ tanks: ₹0.5–0.8 L
- Incremental R&D: ₹0.15–0.3 L | Certification: ~₹0.05 L
- Shared SG&A + marketing + dealer allocation: ₹0.7–1.2 L
- Warranty/service: ₹0.2–0.3 L
- Total cost before profit: ~₹6.1–9.65 L/unit
- Indicative price incl. margin: ~₹6.5–10.7 L/unit

The FCEV is treated as an extension of Suzuki’s existing EV platform, not a standalone manufacturing program — existing BIW, motor, battery, electronics, R&D, logistics and dealer infrastructure are reused.

- Incremental capex is therefore limited mainly to the hydrogen ecosystem: pilot ~₹40–65 L vs full-scale ~₹650–1,050 Cr.
- Cost per unit: pilot ~₹22–35 L vs full-scale ~₹6.1–9.65 L (pre-margin). The main incremental cost drivers remain the fuel-cell stack and H₂ tanks.
- Note: all figures are rough guesstimates for directional analysis only and should be treated as assumptions, not quotable figures.

ANNEXURE 11 – HYDROGEN INFRASTRUCTURE COSTS

11

STAGE 1 – New CHG module integrated into an existing CNG station (350 bar, 50–100 kg/day)

Stage	Scope	Indicative cost (₹)	Confidence
1. Planning, regulatory & site assessment	HAZOP + risk assessment (PESO-mandatory), hazardous-area classification, 500 m layout study, Form H licensing	₹22–45 lakh	GUESSTIMATE (HAZOP requirement itself is REGULATORY-confirmed; cost of the study is not)
2. Civil & structural works	Foundations, protective barriers/bollards, canopy, site modification	₹20–40 lakh	GUESSTIMATE
3. Core H ₂ equipment procurement & installation	Unloading skid, compressor package, cascade storage (100–200 kg), priority panel, 350-bar dispenser, piping/valves	₹2.0–3.9 crore	GUESSTIMATE
4. Safety & electrical systems	H ₂ /flame detection, ESD/interlocks, Ex-rated electrical, earthing	₹35–75 lakh	GUESSTIMATE
5. Controls & integration	PLC/SCADA tie-in with existing CNG site controls	₹10–20 lakh	GUESSTIMATE
6. Testing, commissioning & PESO final inspection	Engineering, certification, commissioning	₹20–35 lakh	GUESSTIMATE

STAGE 2 – Retrofit an existing 350-bar CHG station with a second compression stage (→700 bar dual-pressure)

Stage	Scope	Indicative cost (₹)	Confidence
1. Engineering assessment & regulatory amendment	Retrofit compatibility check on existing cascade/compressor; PESO license amendment (layout change + incremental HAZOP)	₹15–30 lakh	GUESSTIMATE
2. Second-stage (booster) compressor	High-pressure booster to push existing 450–550 bar output to 700+ bar	₹40–80 lakh	GUESSTIMATE
3. 700-bar cascade storage	Additional high-pressure cascade bank	₹30–70 lakh	GUESSTIMATE
4. Pre-cooling/chiller system	Mandatory for 700-bar fast-fill per SAE J2601	₹30–60 lakh	GUESSTIMATE
5. 700-bar dispenser & nozzle	New dispenser, higher-pressure valves/piping	₹30–60 lakh	GUESSTIMATE
6. Controls, piping & commissioning	Additional instrumentation, piping tie-ins, installation, testing	₹30–60 lakh	GUESSTIMATE

STAGE 3 – In House Electrolyser (GUESSTIMATE) - ₹1.8-4.4Cr, maintenance ₹6-22 LPA

SOURCE HIERARCHY USED IN THIS SUBMISSION

Level	Used for
1 — Official problem statement	Scope, constraints, judging criteria
2 — MATLAB sizing tool	Vehicle sizing, energy, tank, stack, power, TCO
3 — SRDI QnA (own research)	Strategy, target customers, safety reasoning
4 — Deployment feasibility study	Infrastructure, manufacturing, localisation
5 — External sourced benchmarks	Global OEM & case-study validation

MODELLING LIMITATION – STATED PLAINLY

HONEST FINDING – MASS-CONVERGENCE LOOP

The corrected buffer-battery mass (37.3 kg, power-limited basis) is computed but not fed back into the vehicle's mass-convergence loop, which still runs on a 20 kg placeholder. This submission uses 37.3 kg throughout. Feeding it back raises total vehicle mass by ~17 kg (~1%) — disclosed, not hidden.

KEY EXTERNAL BENCHMARKS CITED

- IEA: global H₂ fleet ≈130,000 vehicles in 2025, +~20% YoY, dominated by trucks/buses in China
- Toyota Mirai ≈1,168 units vs Hyundai ≈6,861 units — fleet-led vs infrastructure-led OEM strategies
- India green-H₂ tender price: ₹279–336/kg (this TCO model conservatively uses ₹400/kg)
- PESO Gas Cylinders (Amendment) Rules 2024 — CHG dispensing licence requirements

Legend: SOURCED = verified against a named public source · REGULATORY = government/standards-body document · GUESSTIMATE = engineering planning estimate, no public source located — verify with a vendor quote before use in a funding submission. No cost, market size, efficiency value, vehicle specification, hydrogen price, infrastructure CAPEX, OEM sales figure or regulatory claim in this submission is fabricated.